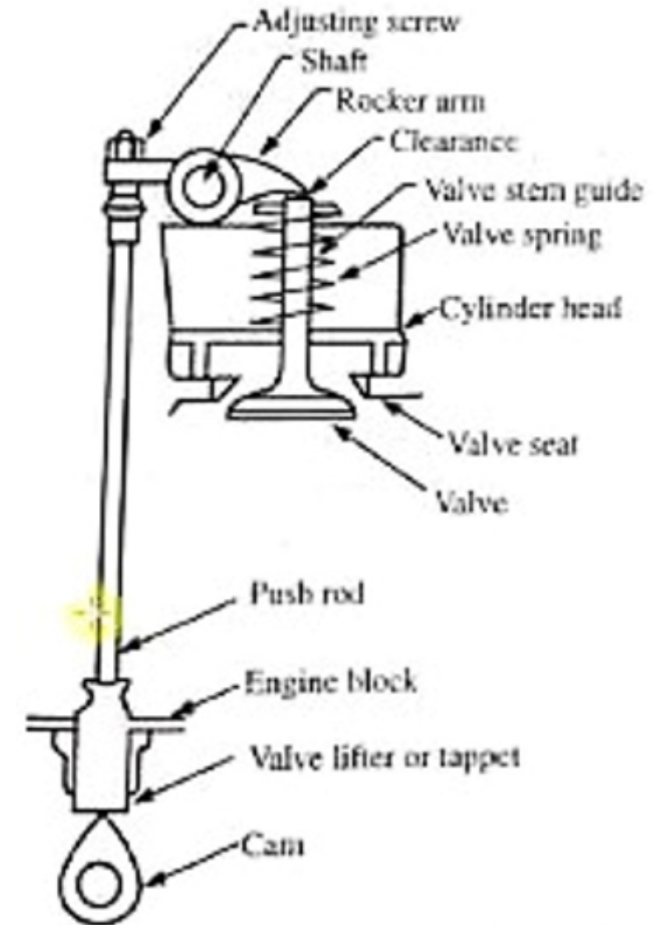


VALVES ACTUATING MECHANISM

Valve actuating mechanism

- A mechanical system that opens and closes a valve
- Within internal combustion (IC) engines, valve actuators translate the camshaft's rotational motion into the linear movement needed to open and close the engine's valves
- These mechanisms are categorized by their power source
- Power sources include manual, electric, pneumatic, and hydraulic systems



VALVES ACTUATING MECHANISM

Common Types of Valve Actuators

1. **Manual Actuators:** These require direct human interaction to operate the valve, often using gears or a lever connected to the valve stem.
2. **Electric Actuators:** Powered by an electric motor, they convert rotational motion into linear or rotary motion to operate various valve types. They can be energy-efficient and are suitable for battery-operated systems.
3. **Pneumatic Actuators:** They use compressed air to generate linear motion, which can then be converted to rotary motion. They are commonly used for remote or automated control.
4. **Hydraulic Actuators:** These actuators use hydraulic fluid pressure to create the necessary force for valve operation, making them ideal for high-force applications or precise control in automated systems.

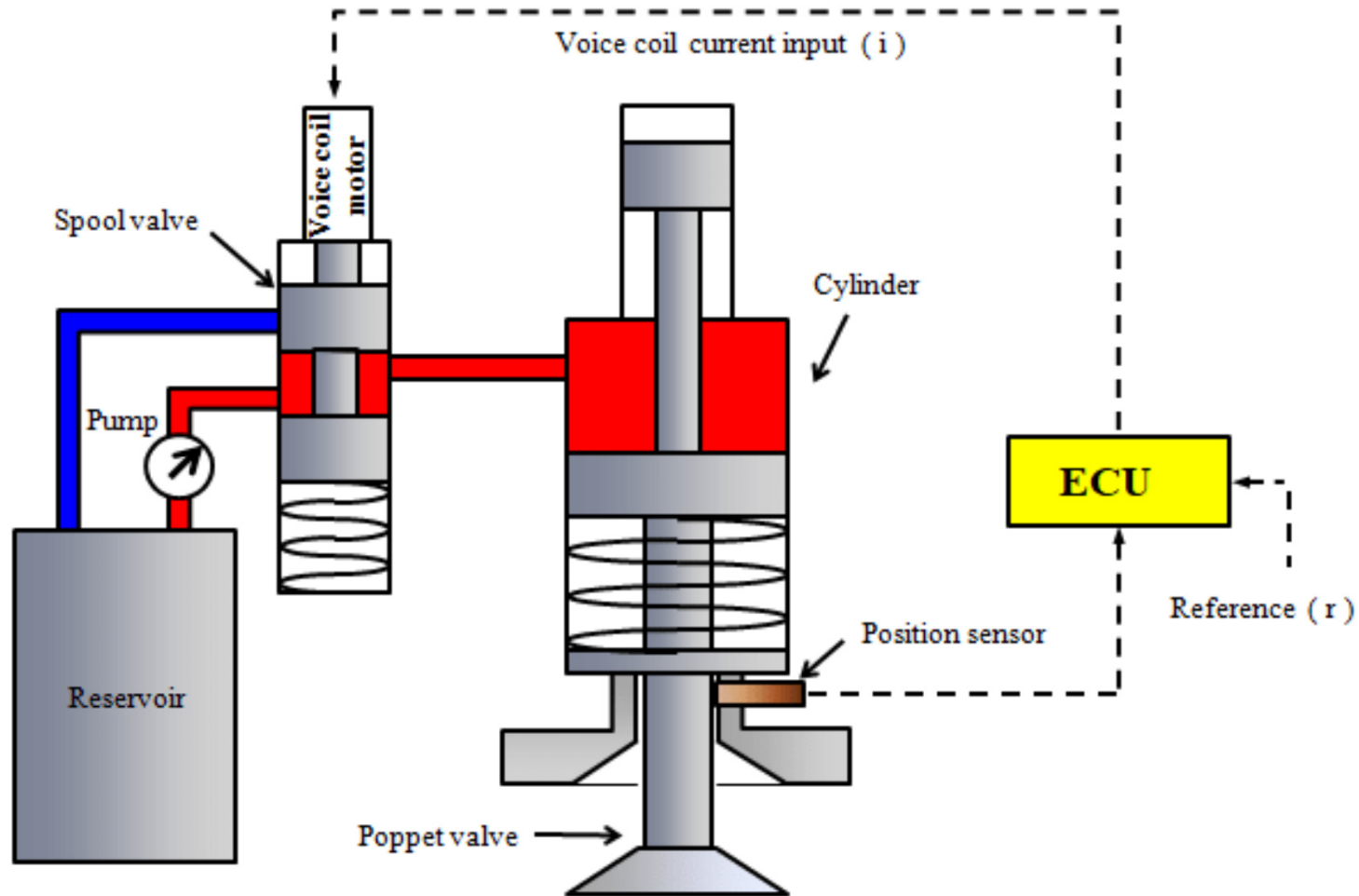
CAM LESS ENGINE VALVE ACTUATION

Cam Less Engine Actuation

- Camless engine actuation uses electronic, hydraulic, or pneumatic actuators to control engine valves instead of a mechanical camshaft
- Allows individual variable valve lift, timing, and duration to significantly improve engine performance, fuel efficiency, and reduce emissions.
- This technology enables infinitely variable control over valve events, which is impossible with traditional camshaft systems.

CAM LESS ENGINE VALVE ACTUATION

Cam Less Engine Working



CAM LESS ENGINE VALVE ACTUATION

Cam Less Engine working

- **Sensors and ECU:** Sensors monitor various engine parameters, such as speed and load. This data is sent to an Electronic Control Unit (ECU), which then sends signals to the actuators.
- **Actuators:** Instead of cams, actuators are placed at the ends of each valve to open and close them precisely and independently.
- **Types of Actuators**
 - **Electromechanical:** Use electromagnets and springs to control the valve's movement.
 - **Electro-hydraulic:** Use hydraulic pressure controlled by solenoids to move pistons that open and close the valves.
 - **Pneumatic/Pneumatic-Hydraulic:** Utilize air or a combination of air and hydraulic fluid to control the valves.

CAM LESS ENGINE VALVE ACTUATION

Benefits

Improved Performance: Allows for optimized valve timing and lift, which can increase power output.

Enhanced Fuel Efficiency: Enables more efficient combustion by precisely controlling valve events for different operating conditions.

Reduced Emissions: More precise control of the combustion process helps to reduce harmful emissions.

Greater Flexibility: Offers infinite variability in valve lift, duration, and timing, providing greater flexibility in engine design.

Simpler Packaging: Eliminates the complex mechanical linkage of the camshaft, potentially leading to a more compact engine design.

CAM LESS ENGINE VALVE ACTUATION

Challenges

Cost: Camless systems are currently more expensive to produce than conventional camshafts.

Power Consumption: The actuators, particularly electromagnetic ones, can consume significant power.

Complexity: The overall system is more complex, requiring advanced electronic controls and reliable actuators.

Noise: The rapid closing of valves and impact on the cylinder head can generate unwanted noise

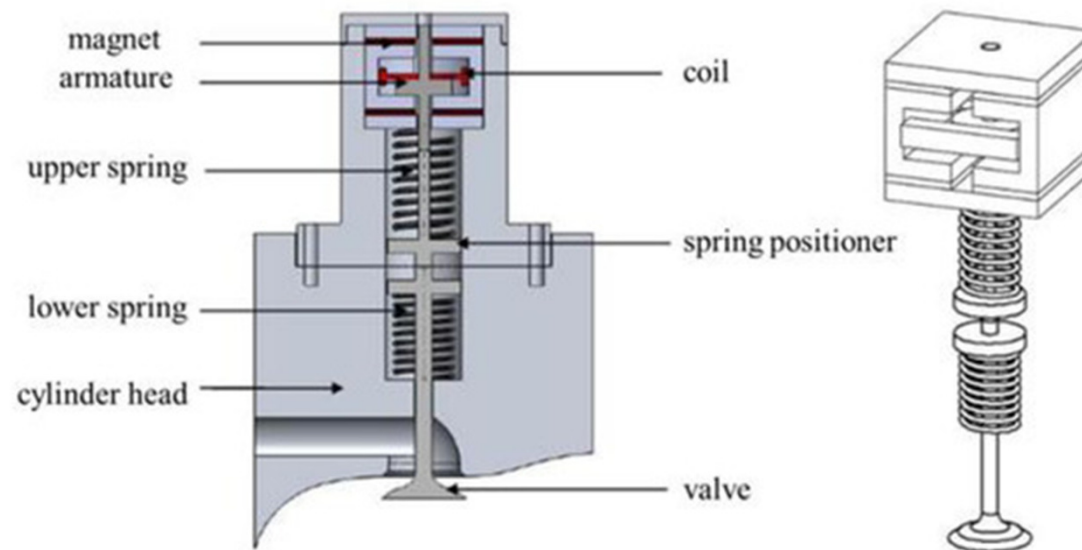
VALVES ACTUATING MECHANISM

Electromagnetic Valves

Electromagnetic valves, or electromechanical valve actuators (EMVAs), are a form of camless valve train system

Instead of a camshaft, these systems use electromagnets and springs to electronically control the opening and closing of the intake and exhaust valves

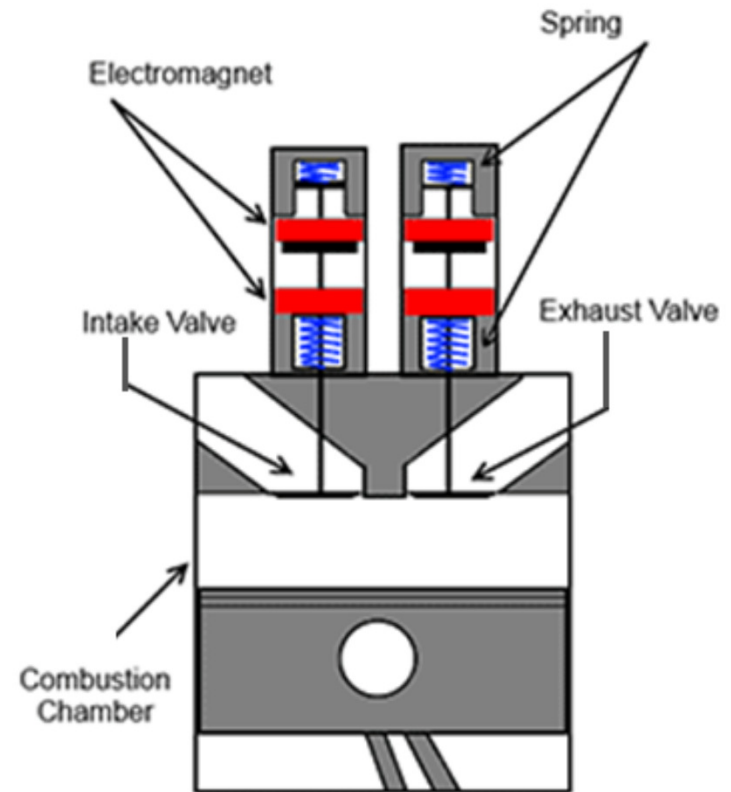
This allows for fully variable valve timing and lift, which can improve engine efficiency and performance.



ELECTROMAGNETIC VALVES

Working of electromagnetic valves

- An EMVA uses a pair of electromagnets and a spring mechanism to actuate the engine valve.
- The armature, which is attached to the valve, is suspended between an upper and lower electromagnet.
- To open or close the valve, an electrical current is sent to one of the electromagnets, which generates a magnetic field.
- The magnetic field attracts the armature, causing the valve to move.
- The spring mechanism provides the initial acceleration, while the electromagnet controls the final movement to ensure a "soft landing" and reduce noise and wear.
- Precise control of the current, often managed by a microprocessor, allows for complete customization of valve timing and lift based on engine conditions.



ELECTROMAGNETIC VALVES

Advantages

Replacing the mechanical camshaft with an EMVA offers several benefits for engine performance and efficiency.

Variable valve control: EMVAs enable fully variable valve timing and lift. This means the timing and duration of the valve openings can be precisely adjusted in response to factors like engine speed and load.

Improved fuel economy: By optimizing valve timing and lift for different operating conditions, EMVAs can reduce engine pumping losses and improve fuel consumption.

Lower emissions: Greater control over the combustion cycle leads to better fuel burning and can significantly reduce harmful exhaust emissions, such as nitrogen oxides (NO_x).

Increased torque and horsepower: The ability to optimize valve timing across the entire operating range allows for greater low-end torque and higher peak horsepower.

No camshaft components: The elimination of mechanical components like the camshaft, timing chain, and associated pulleys reduces engine complexity and friction.

ELECTROMAGNETIC VALVES

Challenges and disadvantages

High power consumption: Repeatedly energizing the powerful electromagnets to open and close the valves at high engine speeds requires significant electrical energy. Some hybrid EMVA designs use permanent magnets to reduce this consumption.

Packaging complexity: The electromagnetic actuators are larger and heavier than traditional valve springs and must be fitted into the already-tight space of the cylinder head, which can compromise volumetric efficiency.

Thermal management: The coils of the electromagnets generate a great deal of heat, which must be dissipated to prevent overheating and component damage.

Precise control and durability: The system requires a complex control unit to manage the rapid and precise valve movements. Ensuring a quiet and soft landing of the valve over millions of cycles without mechanical wear is a major challenge.

Cost: The technology is significantly more expensive to manufacture and implement than a traditional cam-driven system.

IC ENGINE VALVES LAY OUT

Valves lay out

- An IC engine valve layout describes the physical arrangement and position of the inlet and exhaust valves within the engine's cylinder head or cylinder block
- It determines how they control gas flow in the combustion chamber.
- Common layouts are classified as L-head, I-head, and F-head, which refer to the shape of the combustion chamber formed by the valve placement.

IC ENGINE VALVES LAY OUT

Aspects of Valve Layout

Valve Location: Valves can be positioned in the cylinder head (overhead valve or "I-head") or in the cylinder block (flat head or "L-head").

Valve Arrangement: The specific placement of the intake and exhaust valves can form different combustion chamber shapes.

Function: Valves regulate the entry of the air-fuel mixture (intake) and the exit of exhaust gases, a critical function for the engine's operation.

Valve Types: The most common type of valve in internal combustion engines is the poppet valve, which seals against a valve seat to control flow.

Valve Timing: While not directly part of the valve layout, the timing diagram shows when these valves open and close in relation to the piston's position to optimize the four-stroke cycle.

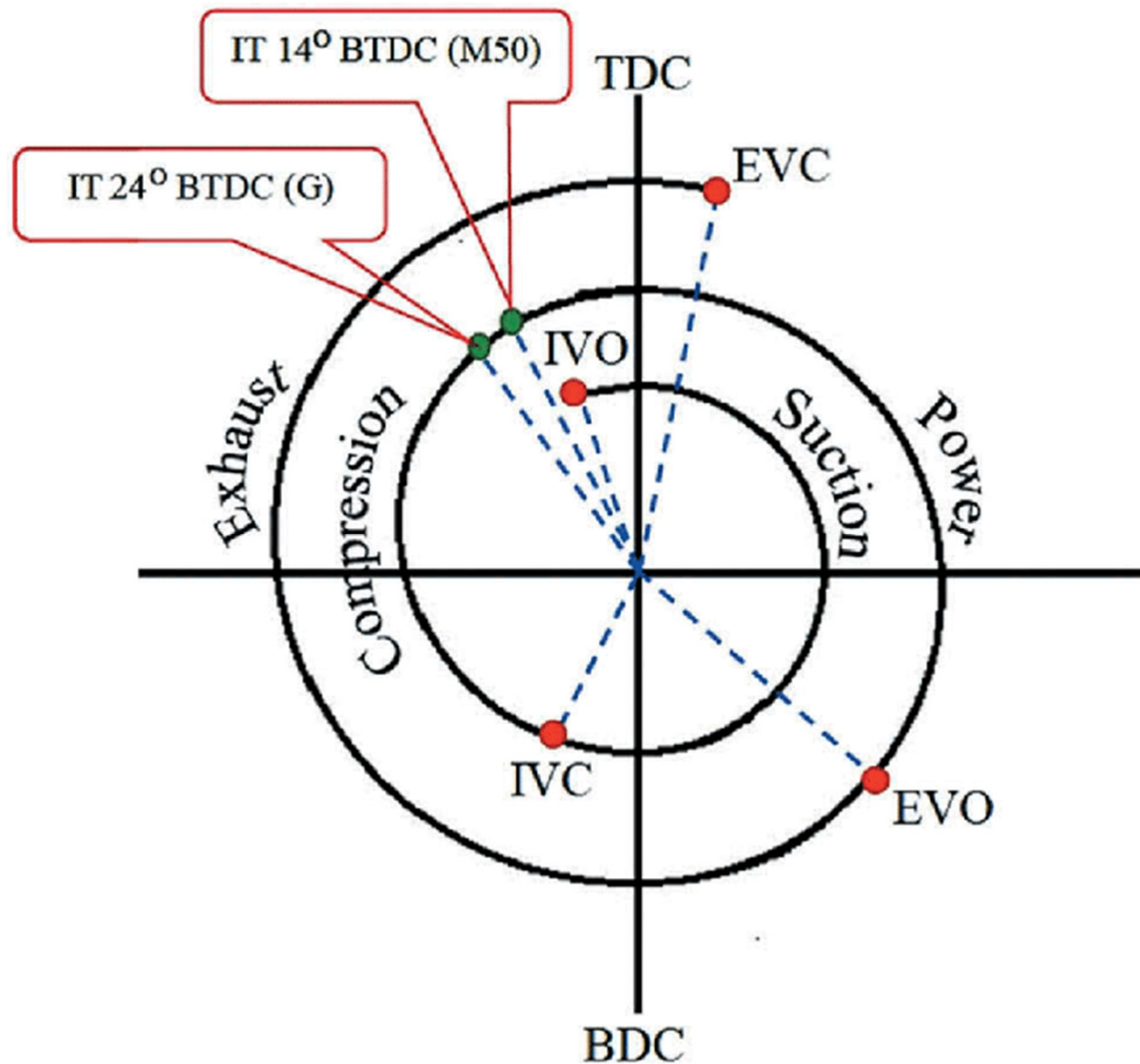
VALVE AND PORT TIMING

Valve And Port Timing

- Valve timing is the precise synchronization of an engine's intake and exhaust valves with the piston's position, measured in crankshaft degrees
- Port timing refers to the opening and closing of ports by the piston in a two-stroke engine
- Both are graphically represented in a valve/port timing diagram
- Its accurate execution is crucial for optimizing engine performance, efficiency, and emissions.

VALVE AND PORT TIMING DIAGRAMS

Valve timing diagram of four-stroke SI engine



IVO – Inlet valve opens

IVC – Inlet valve closes

EVO – Exhaust valve opens

EVC – Exhaust valve closes

IS – Ignition Starts

VALVE AND PORT TIMING DIAGRAMS

Valve timing diagram of four-stroke SI engine

- On an M50 engine, 14° BTDC refers to the ignition timing setting
- The spark plugs are set to fire 14° of crankshaft rotation Before Top Dead Centre.
- This is a performance modification that advances the timing
- 24° BTDC (g) means the engine's ignition timing is set to fire the spark plug 24° degrees Before Top Dead Centre in gasoline engines.
- **M50**
- The BMW M50 is a 1990s straight-6, dual overhead camshaft (DOHC) petrol engine
- It has robust, smooth power, and reliability.

VALVE AND PORT TIMING DIAGRAMS

Need for 14° BTDC advance timing

Performance increase:

- Provides a slight increase in power and torque, especially at higher RPMs.

Optimizing combustion:

- An advanced spark gives the combustion process more time to complete.
- The piston starts its downward power stroke and the expanding gases creates maximum pressure
- Leads to more efficient use of the fuel and a stronger downward push

VALVE AND PORT TIMING DIAGRAMS

Limitations of advanced Ignition timing

Risk of pre-ignition:

- Advancing the timing too much can cause the fuel mixture to ignite too early
- Results in an uncontrolled explosion called "knock" or "detonation," which can severely damage the engine.

Need of Higher octane fuel:

- The risk of knock is why this modification often requires the use of higher-octane fuel.
- Higher octane fuel is more stable and less prone to premature combustion under pressure.

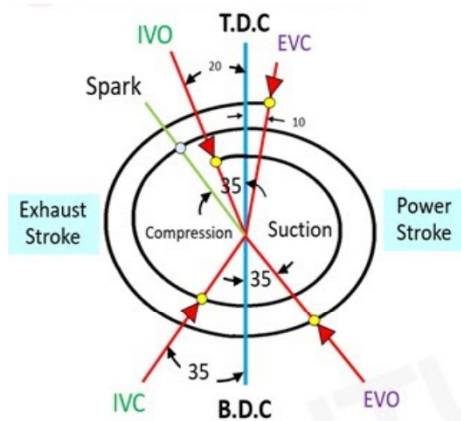
VALVE AND PORT TIMING DIAGRAMS

Octane Number

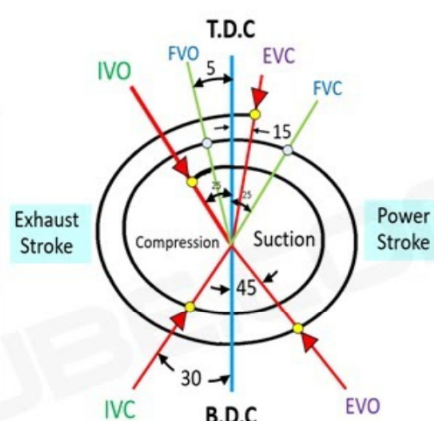
- The octane number measures a fuel's resistance to "knocking" or premature ignition in an internal combustion engine
- It's a rating system where iso-octane, a highly resistant compound, has an octane number of 100, and n-heptane, a prone compound, has an octane number of 0.
- Higher octane numbers indicate greater resistance to knocking
- It allows for higher engine compression and potentially greater efficiency.

VALVE AND PORT TIMING DIAGRAMS

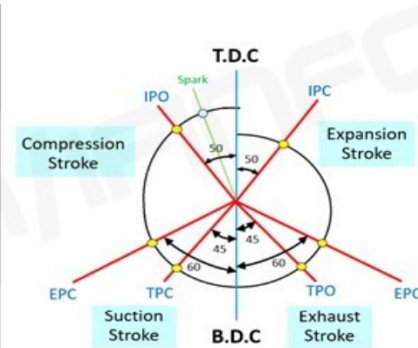
Valve timing diagram of SI and CI engine



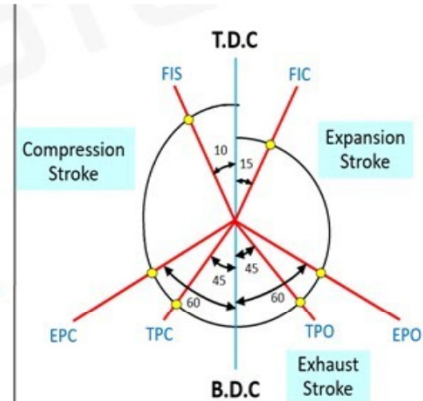
**Four Stroke
Petrol Engine [S.I.]**



**Four Stroke
Diesel Engine [C.I.]**



**Two Stroke
Petrol Engine
[S.I.]**



**Two Stroke
Diesel Engine
[C.I.]**

VARIABLE VALVE TIMING (VVT)

Variable Valve Timing

- Adjusts the opening and closing times of an engine's valves to optimize its performance across different engine speeds and loads.
- It shifts the timing of valve events
- VVT improves fuel economy, reduces emissions, and enhances engine torque and power
- This is achieved by altering the phase angle of the camshaft relative to the crankshaft
- Often using oil pressure controlled by the ECM

METHOD USED TO EFFECT VARIABLE VALVE TIMING

Method used to effect variable Valve Timing

- The primary techniques include cam phasing, cam switching, and camless systems.

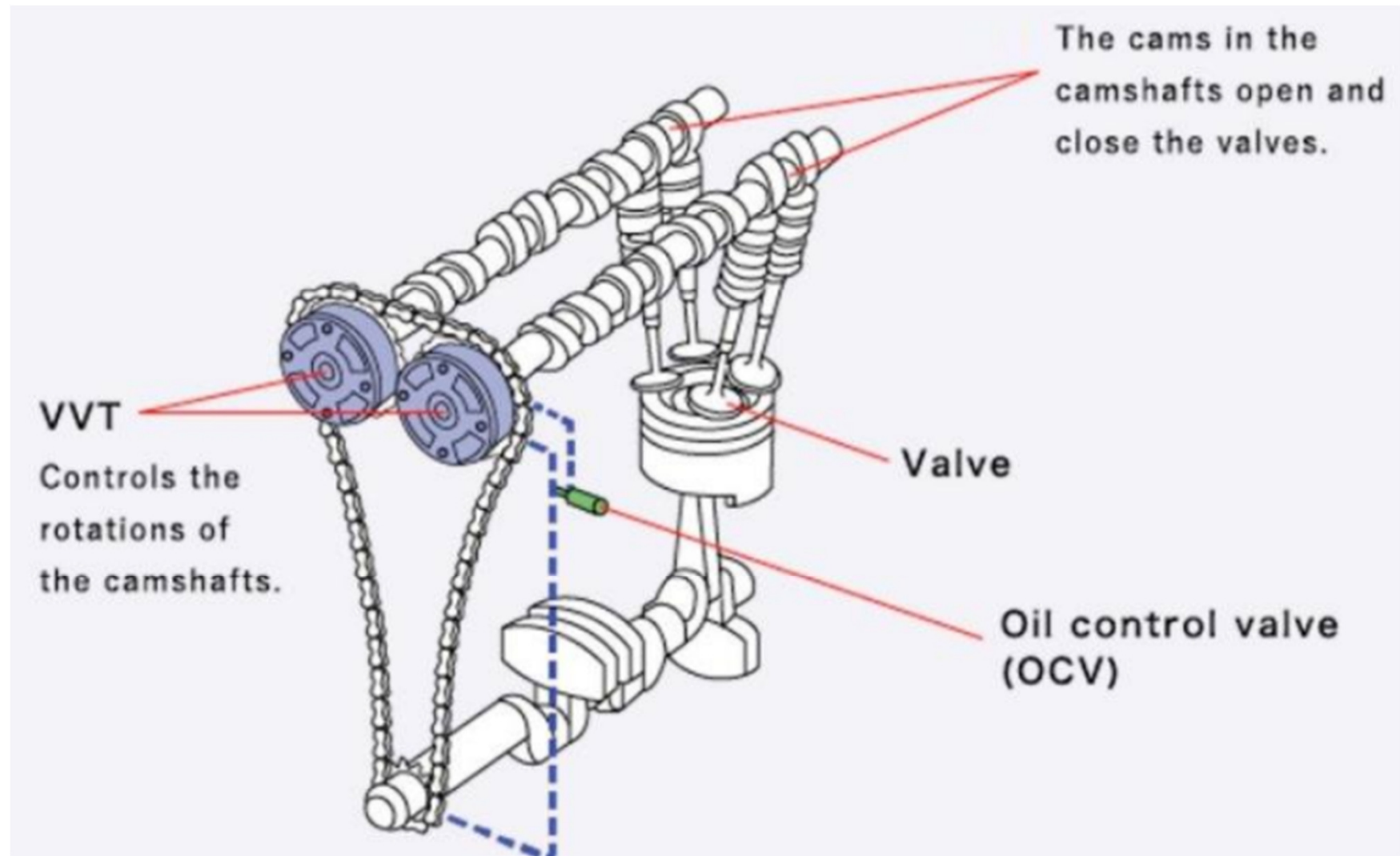
1. Cam phasing

- This is the most common and cost-effective VVT method
- Works by changing the rotational position (phase angle) of the camshaft relative to the crankshaft.

Mechanism:

- A device called a phaser or variator is mounted on the end of the camshaft.
- It consists of an outer sprocket driven by the timing chain and an inner rotor connected to the camshaft.

VARIABLE VALVE TIMING (VVT)



METHOD USED TO EFFECT VARIABLE VALVE TIMING

Working

- The Engine Control Unit (ECU) directs engine oil pressure, regulated by a solenoid, to the phaser.
- The oil pressure moves internal vanes, causing the inner rotor and camshaft to rotate slightly relative to the outer sprocket.
- This advances or retards valve timing.
- Valve opening and closing events happen earlier or later
- The duration and maximum lift of the valves remain unchanged.

Variants:

Single VVT: Controls only the intake or exhaust camshaft.

Dual VVT: Controls both the intake and exhaust camshafts independently for more precise timing adjustments.

METHOD USED TO EFFECT VARIABLE VALVE TIMING

2. Cam switching

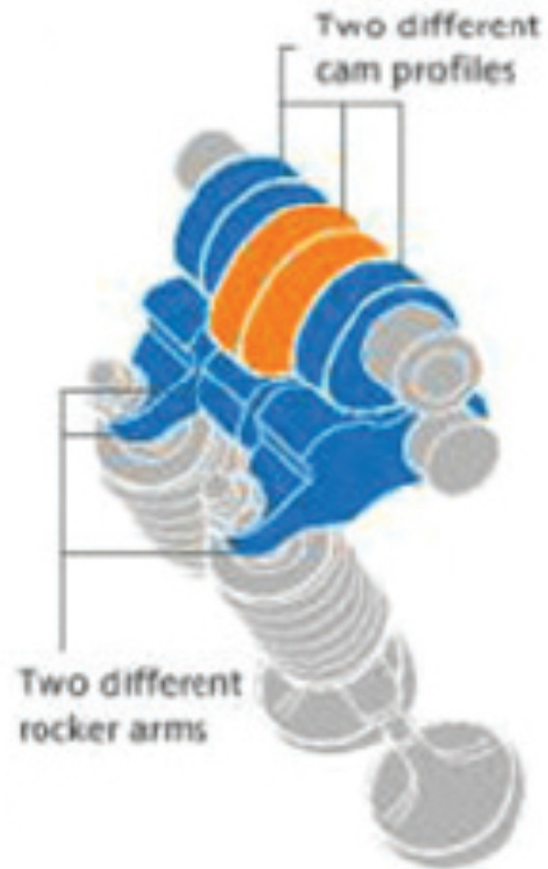
- Uses multiple cam lobe profiles on the same camshaft and switches between them at specific engine speeds.
- This is a more complex and performance-oriented approach.

Working

- A rocker arm follows the cam lobes
- At low RPM, a cam lobe optimized for efficiency and low emissions and controls the valves.
- When the ECU signals the system, typically at high RPM, a hydraulically activated pin locks an adjacent high-lift, long-duration cam lobe for increased power.
- This causes the valve to follow a more aggressive cam profile.
- Valve lift and duration can be significantly altered, providing a noticeable boost in power.

METHOD USED TO EFFECT VARIABLE VALVE TIMING

2. Cam switching



METHOD USED TO EFFECT VARIABLE VALVE TIMING

3. Cam-phasing and cam-switching combinations

- Some advanced systems combine both cam phasing and cam switching to achieve the benefits of both

Working

- These systems use cam phasers for continuous timing adjustment and a separate mechanism for switching between different cam profiles.
- At low speeds, the system uses a fuel-efficient cam profile, with cam phasing adjusting timing.
- At high speeds, the cam profile switches to one with higher lift and longer duration, and cam phasing continues to optimize timing.
- Example: Toyota's VVTL-i system, an evolution of VVT-i, included variable valve lift and duration in addition to continuous timing.

METHOD USED TO EFFECT VARIABLE VALVE TIMING

4. Continuous valve lift and timing

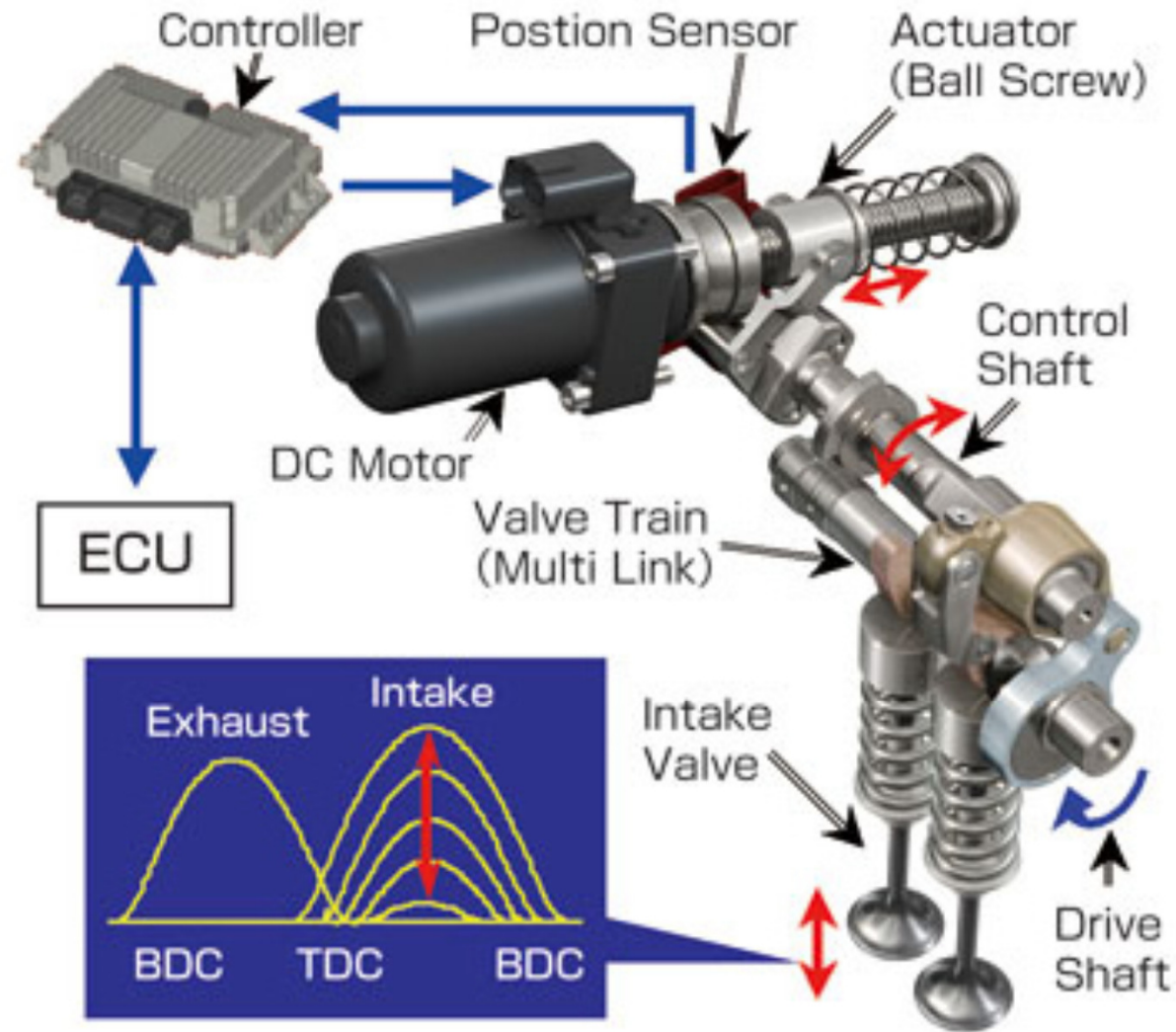
- These systems allow for truly continuous adjustment of valve lift and timing, providing the greatest flexibility but with high complexity.

Working

- Some designs use an oscillating cam, which changes the lift and duration of the valves.
- In these systems, lift and duration are interdependent.
- The ECU controls an electric motor that rotates the eccentric shaft.
- This, in turn, changes the pivot point of an intermediate lever that activates the valves, allowing for infinitely variable lift.
- The mechanism reduces pumping losses and significantly improves fuel efficiency
- BMW's Valvetronic system uses an eccentric shaft with an intermediate lever to adjust intake valve lift continuously

METHOD USED TO EFFECT VARIABLE VALVE TIMING

4. Continuous valve lift and timing



ENGINE TUNING

Engine tuning

- Process of modifying or optimizing an engine's mechanical or electronic components to enhance its performance, fuel economy, or durability.
- Modern engines are often tuned by reprogramming the engine control unit (ECU) through software (ECU remap) or by adding plug-in tuning hardware.
- Tuning can also involve mechanical modifications, such as adding turbochargers or superchargers, to increase the amount of air entering the engine.

ENGINE TUNING PROCEDURE

ECU Reprogramming (Remapping):

- This involves changing the software within the vehicle's ECU to alter parameters like the air-fuel mixture, ignition timing, and boost pressure.
- The ECU receives information from various sensors to ensure optimal engine operation, and tuning modifies these parameters to achieve desired outcomes.

Chip Tuning:

- This involves adding physical "chips" to the ECU
- Also uses plug-in devices that intercept and modify signals sent to and from the ECU.

Mechanical Modifications:

- **Forced Induction:** Installing a turbocharger or supercharger forces more air into the engine's cylinders, allowing for more air fuel mixture to be burned and increasing power.
- **Intake and Exhaust Modifications:** Modifying the air intake to be less restrictive can improve airflow into the engine

ENGINE TUNING PROCEDURE

Benefits of Engine Tuning

Increased Power and Torque: More air and fuel can be burned, leading to greater power output and torque.

Improved Driveability: The engine can be made more responsive, providing smoother power delivery.

Enhanced Fuel Economy: On some engines, particularly diesels, tuning can lead to better fuel efficiency by optimizing the engine's operation

Risks Factors

Factory Compromise: Manufacturers often tune engines to a compromise, balancing performance, reliability, and emissions for a wide range of conditions.

Engine Damage: Improper or poorly executed tuning can push the engine beyond its design limits, leading to premature failures like warped cylinder heads, engine knocking, or sudden engine failure.

Professional Expertise: Tuning, especially involving significant hardware changes, requires expertise to ensure safe and effective results

ENERGY BALANCE SHEET

Energy balance sheet

- An internal combustion (IC) engine heat balance sheet is a chart that tracks the distribution of the total heat energy
- Energy supplied by the fuel converted into useful work, heat absorbed by the cooling system, heat carried away by the exhaust gases, and unaccounted losses, such as radiation and friction.
- This analysis helps determine the engine's thermal performance and identify areas for efficiency improvement.
- The sheet operates on the principle that input heat equals the sum of output heat and heat losses.

COMPONENTS OF AN IC ENGINE ENERGY BALANCE SHEET

Heat Supplied: This is the total heat energy from the fuel, calculated as the mass of fuel consumed multiplied by its calorific value.

Heat Equivalent to Brake Power: This represents the portion of the fuel energy converted into useful mechanical work delivered by the engine's crankshaft.

Heat Absorbed by Cooling Water: The energy transferred to the cooling system to prevent overheating, measured by the flow rate and temperature change of the cooling water.

Heat Carried Away by Exhaust Gases: The heat energy lost through the exhaust system, determined by the mass, specific heat, and temperature of the exhaust gases.

Unaccounted Heat Losses: This category includes all remaining energy not accounted for by the measured outputs, such as radiant heat and friction within the engine.

PURPOSE OF THE HEAT BALANCE SHEET

Performance Analysis: It provides a complete understanding of how the engine utilizes heat energy and where losses occur.

Efficiency Improvement: By identifying the major heat loss points, engineers can design and modify engines to improve overall efficiency and power output.

Data for Engine Design: The sheet offers critical data that helps in the design and optimization of engine components and systems.

Heat Balance Sheet			
Heat Supplied KJ/min	%	Heat Utilized KJ/min	%
Heat supplied by combustion of fuel (Q_s) = 4400	100	1) Heat converted in brake power ($Q_{B.P}$) = 1080	24.5
		2) Heat carried away by cooling water (Q_w) = 1203.76	27.3
		3) Heat carried away by exhaust gases (Q_g) = 154	3.5
		4) Unaccounted Heat = 1962.24	44.5
Total = (Q_s) = 4400	100	Total = (Q_s) = 4400	100

ENGINE COOLING CURVE

Engine cooling curve

- An IC engine cooling curve is a graph that tracks the temperature of the engine as a function of time.
- By plotting the temperature decline once the engine has been switched off, engineers can evaluate the efficiency and characteristics of the engine's cooling system.
- This data helps guide the design of cooling components like radiators and fans to ensure the engine operates within its optimal thermal range.

ENGINE COOLING CURVE

Elements of a cooling curve

- **Cooling rate:** The slope of the curve at any given point indicates the rate at which the engine is losing heat. A steeper slope means faster cooling.
- **Time to reach ambient temperature:** The curve shows how long it takes for the engine to cool down to the surrounding air temperature. This is influenced by the materials used and the efficiency of the cooling system.
- **Thermal management effectiveness:** A cooling curve helps determine the thermal performance of a vehicle's engine under different operating conditions.
- **System limitations:** The curve can help identify which vehicle conditions are most demanding for the cooling system. For instance, testing a vehicle in off-road conditions with a heavy load will reveal the most limiting scenarios for the cooling system.

IMPORTANT OF COOLING CURVES

Optimize thermal management: The curves provide data to design efficient cooling systems that reduce energy waste and lower emissions.

Prevent overheating: The data helps ensure that the engine's cooling system can handle the heat rejection requirements to prevent overheating and premature component failure.

Improve engine performance: An engine operates best within a specific temperature range. Analysing cooling curves helps engineers design systems that reach and maintain this optimal temperature.

Model thermal behaviour: Complex computer models are built using cooling curve data to predict and simulate engine thermal behaviour in various conditions

ENGINE COOLING CURVE

Engine cooling curve

